

Project Proposal

Telco Customer Churn Prediction



Predicting customer churn for better retention and stronger relationships.



COURSE : Introduction to Machine Learning



PROJECT TITLE : Telco Customer Churn Prediction



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1. Problem Statement

Customer retention is important for telecommunication companies. The purpose of this project is to use Machine Learning techniques to predict which customers are likely to leave the company's services.

2. Dataset

We will use the Telco Customer Churn dataset from Kaggle. It contains customer information, including contract type, tenure, internet services, monthly charges, and total charges.

3. Target Variable

The target variable is **Churn**, which has two classes:

- **Yes:** The customer has left the service.
- **No:** The customer continues to use the service.

This is a **binary classification** problem because the model predicts one of two possible outcomes.

4. Data Preprocessing

We will check for missing values, convert **TotalCharges** to numeric format, encode categorical features, and scale numerical features. We will use a scikit-learn Pipeline and ColumnTransformer to prevent data leakage during preprocessing.

5. Exploratory Data Analysis (EDA)

We will create three visualizations: customer churn distribution, customer churn by contract type, and monthly charges by churn status. These visualizations will help us explore patterns and relationships in customer behavior.

6. Evaluation

We will split the dataset into training and testing sets, using 80% of the data for training and 20% for testing. Model performance will be evaluated using suitable classification metrics, including accuracy, precision, recall, and F1-score.